

Code No: 44022

R07

Set No. 2

II B.Tech II Semester Supplementary Examinations, February 2010

SOFTWARE ENGINEERING

**Common to Information Technology, Computer Science And Engineering,
Computer Science And Systems Engineering**

Time: 3 hours

Max Marks: 80

**Answer any FIVE Questions
All Questions carry equal marks**

1. Describe in detail
 - (a) Validation testing
 - (b) System testing [7+9]
2. (a) Explain size oriented metric with an example.
 - (b) Explain function oriented metric with an example. [8+8]
3. (a) Explain the classical life-cycle process model with a real-time scenario.
 - (b) Give the reasons and situations where this model fails. [6+10]
4. What are the major elements of the design model? Explain the abstraction dimensions and process dimensions of the analysis model and design model. [16]
5. (a) Briefly explain about Object oriented systems developments life cycle.
 - (b) Describe state, behavior, and identity with respect to an object with relevant example. [8+8]
6. (a) Explain activities of software.
 - (b) Discuss about cost impact of software defects. [8+8]
7. Discuss an example system where social and political factors may strongly influence the system requirements. Explain why these factors are important in it. [16]
8. (a) What is a legacy software?
 - (b) What type of changes is made to legacy systems if it exhibits poor quality?
 - (c) Explain why legacy systems evolve as time passes. [3+5+8]

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1. Write a short notes on:
 - (a) Data-flow model.
 - (b) State-Machine model. [8+8]
2. Explain the process of refining the architecture into components with the help of a neat diagram. [16]
3.
 - (a) Define the terms software and work product.
 - (b) Explain the impact of software on the society with a real time example. [6+10]
4.
 - (a) Explain the determinants of software quality.
 - (b) Explain public metrics, private metrics. [8+8]
5. Write a note on integration testing & Explain how integration test documentation is done. [16]
6. Write short notes on:
 - (a) quality
 - (b) quality control
 - (c) review meeting
 - (d) review guidelines [16]
7. How the following factors are addressed in interface design?
 - (a) Help facilities
 - (b) Response time
 - (c) Error handling
 - (d) Internationalization [4+4+4+4]
8.
 - (a) Explain in detail the structures language specifications.
 - (b) Draw and explain the sequence diagram representing various consequences during ithdraw from ATM system. [8+8]

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1. Write a short note on:
 - (a) Validation test criteria
 - (b) Alpha & Beta testing
 - (c) Recovery testing
 - (d) Security testing [4+4+4+4]
2.
 - (a) What are the goals of requirements engineering process?
 - (b) With the help of a neat diagram depict the requirement engineering process activities.
 - (c) Explain the spiral model of requirements engineering process. [2+6+8]
3. What is CMMI? List and explain various Key Process Areas assessed by CMMI? [16]
4. Explain about
 - (a) The review meeting
 - (b) Review reporting & Record keeping
 - (c) Review guidelines
 - (d) Sample driven reviews [16]
5.
 - (a) Explain different categories of risks.
 - (b) Discuss about risk identification & risk item checklist. [8+8]
6.
 - (a) Explain why the spiral development is called as a risk-driven process model generator.
 - (b) What are the merits and demerits of the spiral model. [10+6]
7.
 - (a) Discuss the relationship between the concept of information hiding as an attribute of effective modularity and the concept of module independence.
 - (b) Explain how the concepts of coupling and software portability related? [8+8]
8.
 - (a)
 - i. Describe the design process in software development.
 - ii. What are the characteristics and criteria for design?

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Set No. 1

- (b) Create three classes linked by associations to represent a student taking courses in a school. Specify appropriate multiplicity as well as labels for the association. If there is more than one reasonable alternative, explain the advantages and disadvantages of each. [6+10]

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Set No. 3

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Max Marks: 80

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1. Explain the context model with the context of an ATM system. [16]
2. What is a process pattern? Describe the template of a process pattern with an example. [16]
3. Describe component level design & user interface design metrics. [16]
4. (a) What is software architecture? Explain about structured chart with an example.
(b) Justify how modular design is an effective design method? [8+8]
5. Explain the issues of risk. [16]
6. (a) Explain the following features of OOPS
 - i. Message Passing
 - ii. Extensibility
 - iii. Delegation
 - iv. Genericity
(b) Explain overloading of new and delete operators by giving examples. [12+4]
7. (a) Explain defect removal amplification with an example.
(b) Explain about sample driven review. [8+8]
8. (a) Discuss the problems of using natural language for defined user and system requirements.
(b) Explain how structuring natural language into forms can help avoiding some of the difficulties of natural language. [8+8]
